

Mechanical Properties of Materials

Unit I - Mechanical Engineering

ME10008

Comprehensive Study Notes

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1 Classification of Engineering Materials

Engineering materials are broadly classified into three main categories:

1.1 Metals and Alloys

- **Ferrous metals:** Iron, Steel, Cast iron
- **Non-ferrous metals:** Aluminum, Copper, Brass, Bronze
- High strength, good conductivity, malleable and ductile

1.2 Ceramics

- Oxides, carbides, nitrides, silicates
- High hardness, brittle, high melting point
- Examples: Glass, Porcelain, Concrete

1.3 Polymers

- Thermoplastics and Thermosets
- Low density, low strength, high ductility
- Examples: Polyethylene, PVC, Nylon

2 Properties of Engineering Materials

2.1 Elasticity

Definition

The ability of a material to resist deformation and return to its original shape and size when the applied force is removed.

Behavior: When small loads are applied, atomic bonds stretch. Upon unloading, the material returns to its initial state. Can be linear or non-linear elastic.

2.2 Plasticity

Definition

The ability of a material to undergo permanent (non-reversible) deformation under applied forces.

Example: A metal rod bent into a new shape exhibits plasticity as permanent changes occur within the material.

2.3 Ductility

Definition

A measure of a material's ability to undergo significant plastic deformation before rupture.

Measurement:

$$\text{Percent Elongation: } \%EL = \frac{L_f - L_o}{L_o} \times 100$$

$$\text{Percent Area Reduction: } \%RA = \frac{A_o - A_f}{A_o} \times 100$$

where L_f = final length, L_o = original length, A_f = final area, A_o = original area.

Note: Larger %EL indicates higher ductility. Ductile materials show cup-cone fracture.

2.4 Brittleness

Definition

When subjected to stress, material breaks with little elastic deformation and without significant plastic deformation.

Characteristics: Sudden fracture, no warning, low energy absorption.

2.5 Strength

Definition

The ability to withstand an applied load without failure or plastic deformation. It is the capacity to resist force or pressure.

2.6 Malleability

Definition

The ability to be hammered, pressed, or rolled into thin sheets without breaking. Deforms under compression.

Most malleable metal: Gold

2.7 Hardness

Definition

Resistance to permanently indenting the surface. Indicates resistance to plastic deformation or cracking in compression.

Measurement: Indentation test using a 10 mm sphere under known force. Smaller indents mean larger hardness.

Scale: Plastics ; Brasses ; Al alloys ; Steels ; Cutting tools ; Diamond

2.8 Toughness

Definition

Energy required to break a unit volume of material. Approximated by area under the stress-strain curve.

- **Brittle fracture:** Only elastic energy (very small toughness)
- **Ductile fracture:** Elastic + plastic energy (large toughness)

Ranking: Ceramics (small) ; Polymers ; Metals (large)

2.9 Resilience

Definition

Ability of a material to store energy in the elastic region. Energy stored best in elastic region.

$$U_r = \int_0^{\varepsilon_y} \sigma d\varepsilon$$

For linear stress-strain: $U_r \approx \frac{1}{2} \sigma_y \varepsilon_y$

2.10 Impact Strength

Definition

Capability to withstand a suddenly applied load, expressed in terms of energy.

Tests: Izod impact test, Charpy impact test

3 Elasticity Concepts

3.1 Engineering Stress (σ)

Tensile stress: $\sigma = \frac{F_t}{A_o}$ (Units: N/m² or MPa)

Compressive stress: $\sigma = \frac{F}{A_o}$

Shear stress: $\tau = \frac{F_s}{A_o}$

where F = applied force, A_o = original cross-sectional area

3.2 Engineering Strain (ε)

Tensile/Axial strain: $\varepsilon = \frac{\delta}{L_o}$

Lateral strain: $\varepsilon_L = \frac{-d}{w_o}$

Shear strain: $\gamma = \tan \theta = \frac{x}{y}$

where δ = change in length, L_o = original length. **Strain is dimensionless.**

3.3 Linear Elastic Constants

3.3.1 Young's Modulus (E)

Hooke's Law

$$\sigma = E\varepsilon$$

Definition: Modulus of elasticity measures resistance to elastic deformation. It is the slope of stress-strain curve in elastic region.

$$E = \frac{\text{Normal Stress}}{\text{Normal Strain}}$$

3.3.2 Shear Modulus or Modulus of Rigidity (G)

$$G = \frac{\text{Shear Stress}}{\text{Shear Strain}} = \frac{\tau}{\gamma}$$

Definition: Ratio of shear stress to corresponding shear strain. Unit: MPa

3.3.3 Bulk Modulus (K)

$$K = \frac{\text{Direct Stress}}{\text{Volumetric Strain}}$$

Definition: For mutually perpendicular equal stresses, ratio of direct stress to volumetric strain. Unit: MPa

3.3.4 Poisson's Ratio (μ)

Definition

The ratio of transverse (lateral) strain to longitudinal (axial) strain.

$$\mu = -\frac{\varepsilon_T}{\varepsilon_L} = -\frac{\text{Lateral Strain}}{\text{Axial Strain}}$$

Dimensionless. Typical values:

- Metals: $\mu \approx 0.33$

- Ceramics: $\mu \approx 0.25$
- Polymers: $\mu \approx 0.40$

3.4 Relationships Between Elastic Constants

$$E = 2G(1 + \mu)$$

$$E = 3K(1 - 2\mu)$$

$$E = \frac{9KG}{3K + G}$$

$$\mu = \frac{3K - 2G}{6K + 2G}$$

3.5 Deflection in Simple Tension

Longitudinal deflection: $\delta = \frac{FL_o}{EA_o}$

Lateral deflection: $\delta_L = -\mu \frac{Fw_o}{EA_o}$

Note: Larger elastic moduli minimize elastic deflection.

3.6 Elastic Strain Recovery

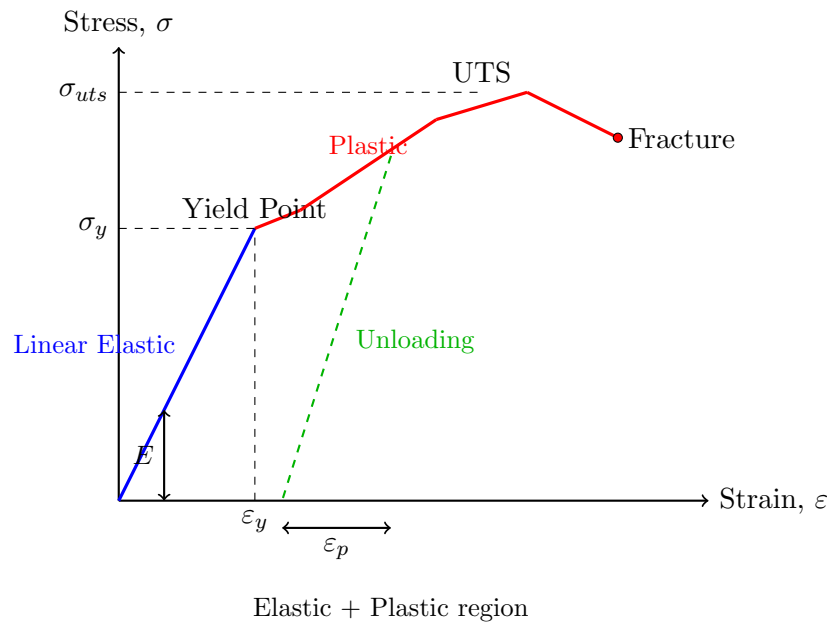
When a material is loaded into the plastic region and then unloaded, only the elastic portion of the strain is recovered. The plastic strain remains permanent.

4 Stress-Strain Curve

4.1 Tensile Testing

Setup: A standard tensile specimen with gauge length is placed in a tensile test machine. An extensometer measures the change in gauge length as load is applied.

4.2 The Stress-Strain Diagram



4.3 Key Points on the Curve

1. **Elastic Region:** Linear relationship between stress and strain. Follows Hooke's Law: $\sigma = E\epsilon$. Material returns to original shape upon unloading. Slope = Young's modulus (E).
2. **Yield Strength (σ_y):** Stress at which noticeable plastic deformation begins. Defined at 0.2% plastic strain offset ($\epsilon_p = 0.002$).
3. **Plastic Region:** Non-linear behavior. Permanent deformation occurs. Material does not return to original shape.
4. **Ultimate Tensile Strength (UTS):** Maximum stress the material can withstand. Peak of the curve.
5. **Necking:** Local reduction in cross-sectional area. Occurs after UTS.
6. **Fracture Point:** Where the material breaks. Ductile materials show cup-cone fracture.

4.4 Plastic Deformation

Plastic strain (ϵ_p): The permanent strain remaining after load removal.

When unloading from the plastic region, the unloading path is parallel to the initial elastic loading line. The horizontal distance from the strain axis to where the unloading line meets the x-axis represents the plastic strain.

4.5 Important Properties from Stress-Strain Curve

- **Modulus of Elasticity (E):** Slope of linear elastic region
- **Yield Strength (σ_y):** Stress at 0.2% offset
- **Ultimate Tensile Strength (UTS):** Maximum stress
- **Ductility:** Total strain at fracture (elongation)

- **Toughness:** Total area under the curve
- **Resilience:** Area under elastic region (triangular area)

5 Special Topics

5.1 Surface Fatigue and Contact Fatigue

Occurs when two surfaces contact each other under pure rolling or rolling with slight sliding. Maximum shear stress exceeds the fatigue limit, causing material spalling (e.g., gear teeth, railroad tracks).

5.2 Comparison of Material Behaviors

Property	Ductile Materials	Brittle Materials
Plastic deformation	Large	Minimal
Fracture	Cup-cone, gradual	Sudden, no warning
Toughness	High (large area)	Low (small area)
Examples	Metals, Polymers	Ceramics, Glass

6 Quick Reference Formulas

Essential Formulas

Stress: $\sigma = \frac{F}{A_o}$, $\tau = \frac{F_s}{A_o}$

Strain: $\varepsilon = \frac{\delta}{L_o}$, $\gamma = \tan \theta$

Hooke's Law: $\sigma = E\varepsilon$

Elastic Constants:

- $E = 2G(1 + \mu) = 3K(1 - 2\mu) = \frac{9KG}{3K + G}$

- $\mu = \frac{3K - 2G}{6K + 2G}$

Deflection: $\delta = \frac{FL_o}{EA_o}$

Ductility: $\%EL = \frac{L_f - L_o}{L_o} \times 100$, $\%RA = \frac{A_o - A_f}{A_o} \times 100$

Resilience: $U_r \approx \frac{1}{2}\sigma_y\varepsilon_y$

7 Exam Tips

Key Concepts to Remember

1. Understand the difference between elastic and plastic deformation
2. Know how to calculate stress, strain, and apply Hooke's Law
3. Memorize relationships between E , G , K , and μ
4. Be able to identify key points on stress-strain curve
5. Understand ductile vs brittle behavior
6. Know how to calculate %EL, %RA, and deflection
7. Toughness = area under entire curve; Resilience = area under elastic region

— End of Unit I Notes —